

3D printed vorticella-kirigami inspired sensors for structural health monitoring in Internet-of-Things

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ARTICLE INFO

Keywords:

Kirigami
Vibration sensor
Structural health monitoring
Internet-of-Things
Vorticella
3D printing

ABSTRACT

Kirigami is an ancient paper art that involves cutting slit strategies, which could provide exotic and controllable engineering properties. Especially, kirigami can be combined with bio-inspired designs to improve its mechanical behaviors. A highly sensitive 3D vibration sensor has been developed using a 3D-printed structure inspired by vorticella-kirigami (VK). VK-inspired kirigami exhibits 2D-to-3D morphing and shape-programmable properties. Specifically, the VK-inspired vibrational structure has been optimized for a frequency of 2 Hz, with high sensitivity of 0.80 pF/ms^{-2} . Structural health monitoring (SHM) is an essential application for ensuring the safety and reliability of civil infrastructure. Therefore, the tools for widely and easily accessible monitoring systems are highly demanded, considering a monitoring system with Internet-of-Things (IoT) as a promising application in this paper. The VK-inspired vibrational sensors can detect the impact delivering from the epicenter on fragile components of the structural structures through the message queuing telemetry transport (MQTT) protocol. Moreover, it allows seamless access to real-time or saved vibration data via multiple smartphones and personal computers. These SHM technologies hold the potential to significantly enhance their adoption in safety-critical domains.

1. Introduction

Structural health monitoring refers to a process of damage identification or anticipation for aerospace, civil, and mechanical engineering. In the process, analyzing the structural vibration is important because it provides a significant clue to addressing structural safety issues [1–3]. Structural vibration occurs when external forces caused by sources such as compressors, pumps, humans, and engines cause deck beams to vibrate. This vibration can have negative impacts on the stability of building structures, such as breakages [1,2] and aging issues [3]. Thus, continuous monitoring of structural vibration is crucial for ensuring the health of structures [4]. However, setting up such a monitoring system can be expensive and challenging [3–6]. To address this issue, there is a need for a simple, portable, and cost-effective vibration sensing system with efficient data acquisition, processing, and monitoring capabilities.

Recently, various vibration sensors have been developed. For instance, a piezoresistive vibration sensor with a microstructured scale has been created using porous poly-dimethyl siloxane (PDMS) with graphene elastomer ink [7] or a graphene-based composite [8,9]. Self-powered vibration sensors have been demonstrated using magnetic

levitation with a helix or spring structure [10] or a triboelectric nano-generator [11]. Vibration detection through capacitive variation has been reported using silver/gold nanowires [12] or other conductive nano- or micro-structures [13]. These sensors have shown reliable performance in detecting even subtle vibrations such as voice [14] or wind turbulence [15]. However, these vibration sensors are manufactured using complex micro- or nano-fabrication processes, making it difficult to apply them to customized structural health monitoring instantly without the additional development of appropriate production and integration methodologies.

In contrast, the architectural vibration sensors alleviate the material preparation and fabrication-related issues as they can show reliable vibration sensing performance without the need for complex micro- or nano-fabrication processes [16]. For instance, reliable vibration sensing performance has been demonstrated using conductive materials such as silver paste or conductive foils, single or double helix structures with copper foils, and conventional spring structures in marine applications and for detecting abnormal vibrations from vehicles [16–19]. The architectural approach to vibration sensors also reduces the cost and complexity of the sensing diaphragm from a material perspective.

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<https://doi.org/10.1016/j.matdes.2023.112332>

Received 24 June 2023; Received in revised form 13 September 2023; Accepted 15 September 2023

Available online 16 September 2023

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Within the realm of architectural approaches, kirigami, an ancient art involving intricate paper cutting, emerges as a promising contender for crafting vibration-sensitive diaphragms. This is due to its capacity to yield a diverse array of structural configurations and mechanical attributes. It has been evolved in design and fabrication to create multifunctional structures of varying scales [20]. Through the introduction of slits onto a two-dimensional (2D) surface, distinct mechanical characteristics such as shape morphing, exceptional stretchability, multistability, and customizable auxetics are induced [21–25]. Additionally, this cutting method has also been applied to bio-inspired structures. Vorticella, a protozoan, is well-known for its inspiration in bio-inspired structures [26]. It has a contractive spiral organelle, a concentric coiled structure, that allows it to stretch [27]. This structure can be created using cutting patterns on a flat 2D sheet. For instance, a three-dimensional (3D) helix structure can be created from the 2D patterned kirigami. This feature makes the 3D structure suitable for use as a vibration-sensitive diaphragm due to its ability to exhibit differences in structural deformation between 2D and 3D shapes [21,28].

Dealing with noise signals unrelated to structural vibration remains a challenge in the development of these vibration sensors. To address this, various filtering methods have been explored, such as the use of low- and high-pass filter circuits with relevant software and signal processing [29]. Signal processing is effective in removing noise signals, including electromagnetic interference, but not physically random vibrations due to ambiguity in sampling [30]. As a result, the self-noise-filtering capability of the sensors is a crucial consideration. For instance, advanced hydrogel molecular dynamics have been reported as a method of self-noise filtering by mechanical damping for improved voice or physiological detection [31]. The molecular dynamics at specific frequencies change the mechanical properties of the hydrogel, resulting in a self-high pass filter. Instead of such advanced molecular dynamics, other types of self-filtering features can be expected from the kirigami structure. The mechanical properties of kirigami can be adjusted by changing its slit pattern design, making it shape-programmable [32–34]. Based on the slit design, the natural frequency of the Kirigami structure can be tuned, resulting in a significant resonance at the desired frequency. From these features, the unwanted physical frequency can be filtered out naturally because except around significant resonance, the range of the other frequencies will be very decent compared to the resonance frequency. The kirigami diaphragm allows specific controls of its natural frequency due to its programmable properties. Traditionally, significant resonance has been considered undesirable due to its potential to lead to failures in sensing diaphragms. Nonetheless, kirigami, owing to its distinctive mechanical properties outlined above, is exempt from such concerns. Thus, more research is needed to tune the natural frequency of the kirigami structure to obtain its optimized harmonic response at specific vibrations.

In this paper, we present a vorticella-kirigami (VK) inspired vibration sensor for structural health monitoring. We first examine the mechanical vibration behaviors of the VK structure and compare it to conventional spring structures. Our study shows the reliable vibration features of VK shape morphing from 2D to 3D. Then, we adjust the VK structure to improve the detection of specific vibrations using its shape-programmable properties. We focus on detecting vibrations related to habitual human behaviors, such as walking, tapping, and knocking [35], which can potentially cause the breakage of fragile components in the structure [36]. Therefore, we have established the optimal criteria for VK structural design to be responsive to vibrations at 2 Hz. This frequency range is intricately linked to human habitual behaviors and holds significance for structural health monitoring applications in our research. Finally, for practical structural health monitoring targeting the internal infrastructure of the building, we demonstrate the ability to monitor vibrations at specific positions within a structure using the IoT-MQTT protocol.

2. Materials and methods

2.1. Fabrication of the VK structure and conductive path on the structure surface

The CAD design was prepared using the Solidworks 2021 software. The designs for the radius and thickness of the VK structure are 7.5 mm and 0.1 mm, respectively. The typical spring design has the same radius and thickness as the VK structure. The turning number of the spiral path for the VK design is 8, which is also used to set the spring layer number. After CAD preparation, the files were exported to STL format. The g-code for the 3D printing was generated using the Ultimaker Cura program (Ultimaker Ltd.). FFF type 3D printer (T-Rex 3.0 multifunctional 3D printer, Vivedino Ltd) was used. The speed, infill, temperature, and layer height for commercial PLA filament printing parameters were set to be 55 mm/s, 100 %, 203 °C, and 0.05 mm, respectively. The bed temperature was set to room ambient temperature. After 3D printing, the samples were carefully transferred to a flat glass plate and inspected. Following the inspection of 3D printed samples, we have verified the consistency in sensor amplitude throughout the calibration processes. We utilized a TPU-Ag composite ink with exceptionally high viscosity, which maintains a liquid-like consistency until it fully dries. The conductive silver paste for the active layer of the VK structure was prepared by dissolving 0.4 g polyurethane into a blend (1 ml N, N-dimethylformamide (DMF) with 4 ml tetrahydrofuran (THF)) by SpeedMixer DAC150.1 FVZ-K (FlackTek, Inc.), followed by adding 2 g silver flake (Inframat Advanced Materials, US). After thoroughly mixing, the viscosity of the silver paste is controlled by heating at 65 °C for the ambient time. Then, the spiral designs for the VK structure are programmed and printed on the surface of the 3D printed VK structure using a direct writing printer SHOT mini 100Sx (Musashi Engineering, Inc.) with a printing speed of 1 mm/s. Also, it has been printed within the valleys along the spiral path of the VK structure, effectively confining the inks within designated areas and enhancing adhesion between layers. After printing, the silver paste on the surface of the VK structure was cured at room temperature for 5 h. Successively, the printed silver paste has been coated with commercial hydrophobic Eco-Surf 00–30 to minimize the effect of ambient moisture and dust.

2.2. Experimental setup to evaluate the vibration sensing performance

The 3D printed shaker machine has been fabricated to evaluate the sensor performance. For the precise controls of imposing vibration, the stepper motor (12 V NEMA 17 stepper motor with cable, Sparkfun electronics) with DRV 8825 stepper motor driver (Pololu, Ltd) is used. Pro Micro – 5 V/16Mhz (Sparkfun electronics) as the microcontroller controls the motor operation. The rotary potentiometer was used to adjust the motor speed. A 12 V, 3A commercial adaptor was used for its power supply. For details on stepper motor operation, please check [Supplementary Fig. 5](#). All parts for the housing shaker electronic devices were designed using Solidworks 2021 and fabricated using an FFF-type 3D printer (Ultimaker, Ltd) with commercial polyethylene terephthalate glycol (PETG) filament. The printing parameters were 55 mm/s, 50 %, 245 °C, 60 °C, and 0.2 mm for printing speed, infill, printing temperature, bed temperature, and layer height, respectively. Also, a commercial accelerometer (ADXL326) with an Arduino system was attached to the sample stage of the shaker to check and monitor the imposing vibration. For applying the identical vibration amplitude of the shaker at different imposing vibrations, six commercial springs at the rod with 3D printed blocks to adjust spring forces are used to push down the oscillating sample stage. For the evaluation of the VK vibration sensor, the sensor was positioned on the sample stage of the shaker and directly connected to the Arduino nano, and measured capacitance changes with 115,200 of the data rate in bits per second (baud) for serial data transmission with 1 ms delay time. For information about the mechanical simulation and 3D printed steel seismic mass, please check

Supplementary Note A.

2.3. Demonstration of structural health monitoring system

After preparing three VK vibration sensing IoT systems, the systems were placed on the side of the glass wall on different bridges. The location of the experiments was the 4th-floor bridge of two different buildings at Simon Fraser University, Canada. After installing and connecting sensors to the MQTT server, knocking impact tests were performed at three different positions of glass walls on two different bridges: 1) at the board-floored building with a light knocking impact and 2) at the concrete-floored building with a higher kicking impact for the demonstration of severe vibration.

3. Results and discussion

3.1. Structural health monitoring system

Fig. 1a shows the overview of the vibration sensing IoT system. We installed three designated VK vibration sensors onto the glass side of the bridge, which is integrated with the message queuing telemetry transport (MQTT) protocol for wide accessibility of real-time monitoring. This system allows the in-situ vibration signals to be subscribed to various portable devices such as smartphones or personal laptops. 3D printing, as a representative of additive manufacturing, is a solution to fabricate such complex structures because it requires one-step processes with minimized waste, as shown in Supplementary Fig. 1. Moreover, the feasible manufacturing of different electronic layers with conducting materials such as silver paste [37] or copper, [38] is available in the 3D structures [37–39]. Therefore, we utilized dual-material 3D printing techniques for low-cost and facile fabrication of VK vibration sensors as shown in Fig. 1b. The vibration-sensitive layer of the VK sensor, the diaphragm, was fabricated by Filament fused fabrication (FFF)-based 3D printing, and its conductive path to prepare the paired electrodes for the

capacitive function was prepared by the direct ink writing (DIW) 3D printing. When the VK diaphragm is vibrated, it shows a 2D to 3D shape morphing behavior [21] as shown in the schematic of Fig. 1c and Supplementary Fig. 2a and b. As shown in Supplementary Video 1, the vibration-sensitive layer is a principal basis of external vibration because of its dimensionally shifting spring behaviors. After the preparation of the sensitive layer, all sensor components are integrated as in Fig. 1d.

3.2. 2D to 3D morphing of VK structure

Fig. 2a describes how the VK is inspired and created as a vibration-sensitive layer. Nature provides a meaningful inspiration to set the design of engineering structures. Vorticella is well-known for its stretchability in nature [26,27]. When the chemical addition of the calcium ion (Ca^{+}) mechanically stimulates it, the organ of the stalk is expanded to be more than a factor of 10. After the removal of the Ca^{+} ion, it returns to its original structure [40]. Within the realm of these intricate motion mechanisms exhibited by vorticella, its structural dynamics have served as a source of inspiration across various engineering domains. A notable instance involves the utilization of its 2D to 3D morphing capabilities to generate spiral 2D shapes that facilitate shape transformation in three dimensions [26,27,40,41]. Furthermore, extending beyond basic 2D configurations, we draw from both the vorticella and kirigami (VK) influences to implement the advantageous aspects of kirigami patterns – defined by repeated cutting slits – that harness the unique structural behaviors inherent in both vorticella and kirigami [20,21]. Also, the vibration behavior of the VK structure was investigated and compared to a conventional 3D spring using the finite element analysis (FEA) tool, Ansys. Two different materials are considered: polylactic acid (PLA) for the VK structure and conductive path along the VK spiral line. Fig. 2b shows the structural deformation and stress distribution. For the computational model of the VK and spring structure, 2 Hz, 0.8 g, and 60 mm/s² are set as vibration frequency,

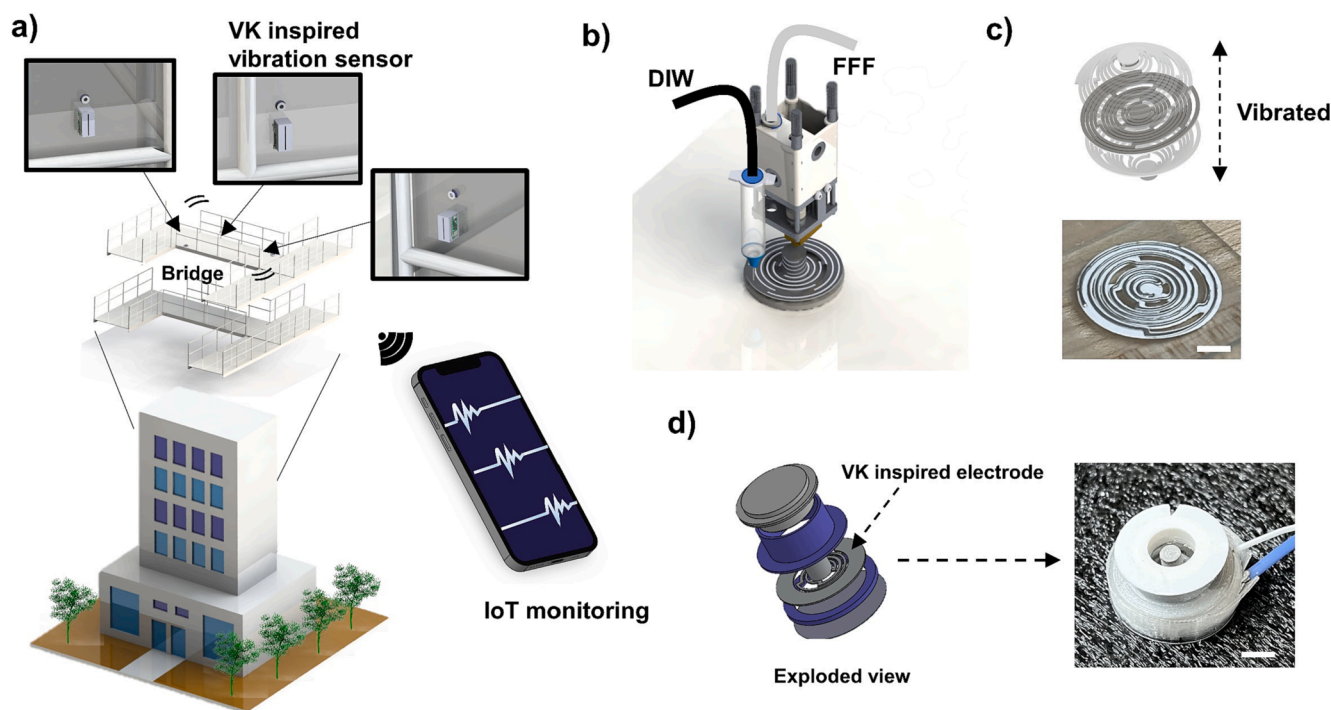


Fig. 1. Structural Health Monitoring System. a) Schematic drawings of structural health monitoring using vibration sensing IoT systems. Three VK vibration sensors have been attached to the glass side of the bridge. When they detect the vibration of the glass, the vibration signals are instantly monitored using portable devices such as smartphones through the IoT-MQTT system. b) FFF- and DIW-3D printers have been used to fabricate the kirigami structure and its conductive path. After fabrication of c) the vibration sensing diaphragm, d) the exterior housing parts of the sensor have been prepared, assembled, and the final vibration sensor has been prepared.

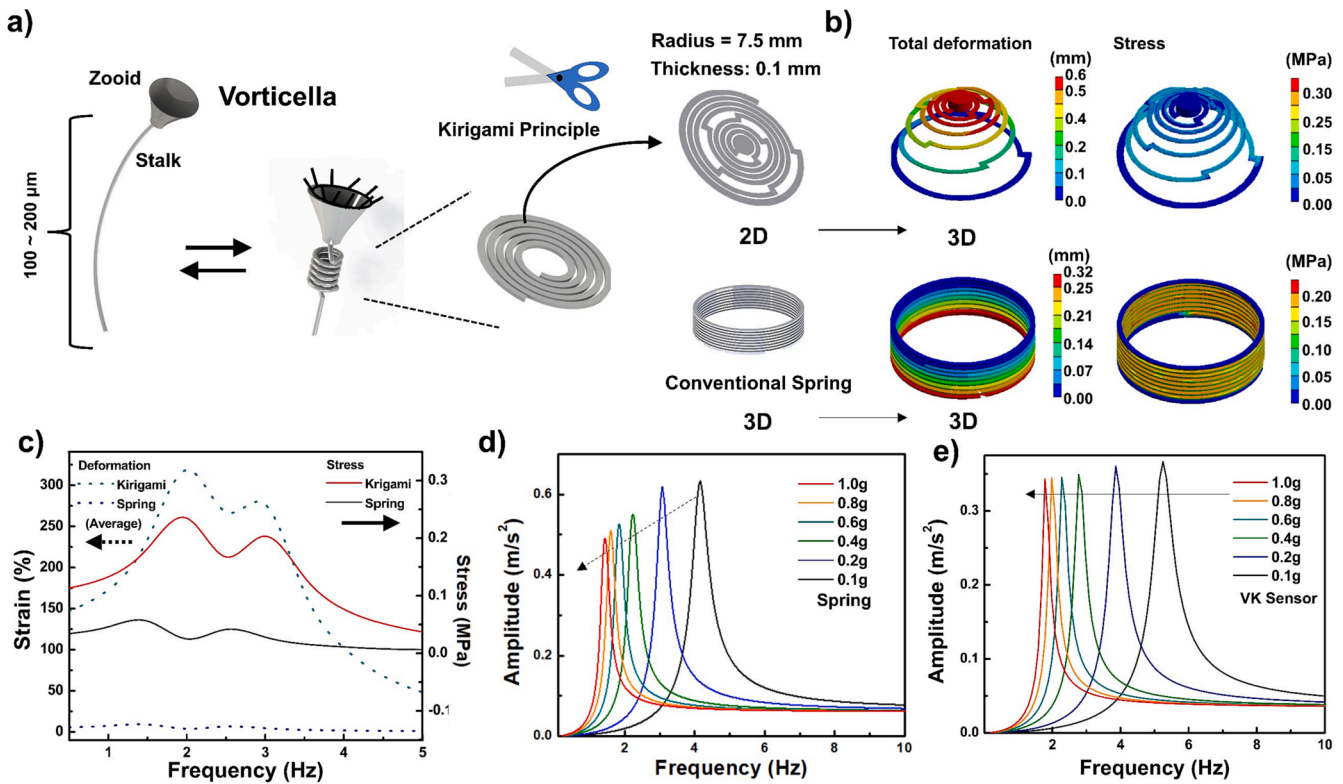


Fig. 2. 2D to 3D Morphing of VK Structure. **a)** Schematic drawings of inspiration and design for the VK structure. The vibration properties of the VK structure are investigated and compared to the typical spring structure. Both structures have the same radius and thickness. **b)** The simulation results of the VK and spring structures at the vibration frequency of 2 Hz and **c)** their corresponding responses at different applied frequencies. Line and dot curves indicate the stress and total deformation of the structures, respectively. Harmonic responses of **d)** the spring and **e)** VK structures with different seismic masses.

seismic mass, and acceleration, respectively. The maximum deformation of the VK structure is 0.6 mm, while the spring structure deforms to its half-length. Also, the minimum stress distribution of the VK structure is mainly observed. The high-stress area was concentrated on only slit edges along the spiral shape. In contrast, the spring structure shows the intensive stress distribution on the entire structure. Therefore, it is foreseeable that the transformation of the VK structure from 2D to 3D would exhibit a greater range of deformation in vibration scenarios compared to the conventional spring structure. Also, their harmonic responses at the broader range of frequency, 0 ~ 5 Hz, are presented in Fig. 2c. The VK structure shows severe deformation at all frequency ranges compared to the spring structure. However, the average stress of the VK structure is about twice higher as that of the spring structure. As shown in Supplementary Fig. 3a, the conductive path along the VK structure has applied stress because of its higher stiffness than PLA. However, the VK structure shows a maximum of 300 % strain, while 25% strain is observed at the usual spring structure. The strain is determined as a percentage by calculating the ratio of the total deformation to the height of the structures. Total deformation is the average value, including in-plane and out-of-the-plane deformation of the VK and spring structure. Therefore, the VK structure shows promising potential for developing a vibration-sensitive diaphragm since its drastic deformation under mechanical loads can provide sensitive capacitance changes as a diaphragm. Moreover, as shown in Fig. 2d, the peak of harmonic response for the spring structure has been shifted from 4.5 Hz to 1.5 Hz when setting higher seismic mass, 0.1 g to 1.0 g. In the case of the VK structure, the peak has been shifted from 5.8 Hz to 2 Hz. The VK structure maintains the responded amplitude in Fig. 2e during its peak shifting, ~ 95 %. However, that of the spring structure is dramatically reduced to 78 %. Additionally, we have summarized the relationship between the seismic mass and the corresponding amplitude in Supplementary Fig. 3c. Thus, the VK structure demonstrates enhanced

suitability for vibration sensor applications focused on specific frequencies. This advantage stems from its minimal amplitude reduction resulting from alterations in the seismic mass. This characteristic suggests the potential for a manageable target-vibration-sensitive layer with reduced requirements for additional calibration processes. In short, the VK structure remains reusable for measuring vibration intensity at targeted frequencies through straightforward replacement of the seismic mass, in contrast to spring structures. Moreover, the damping behavior of the VK structure is considered to support its genuine behaviors, which is described in Supplementary Note B.

3.3. Shape programmable properties of the selected VK structure

The mechanical properties of kirigami structures vary depending on both the shape and cutting patterns, which is generally referred to as “shape programmability” in related studies. This shape programmability is also present in the VK structure, meaning that its vibration response behavior can be altered as the spring constant or structural stiffness is modified by changing the slit patterns [21–24]. The selection of the parameter for the VK structure is determined by the detection of specific external vibrations related to unintentional human behaviors. Thus, the parameter selection for the design of the VK structure aims to maximize harmonic response at 2 Hz of the frequency and its full-width height maximum (FWHM). Fig. 3a shows variable geometrical parameters in the design of the VK structure. From the variable parameters, various shapes of the VK structure are generated. As the parameters, α angle for cutting slits and turning numbers of spiral curves are selected as variable parameters. Also, seismic mass design is considered as a variable parameter because the vibration of the structure is significantly affected by the area of the seismic mass [42]. Accordingly, three different mass elements are considered for the VK structure as shown in Fig. 3b. Initially, the harmonic responses of various angular VK designs at

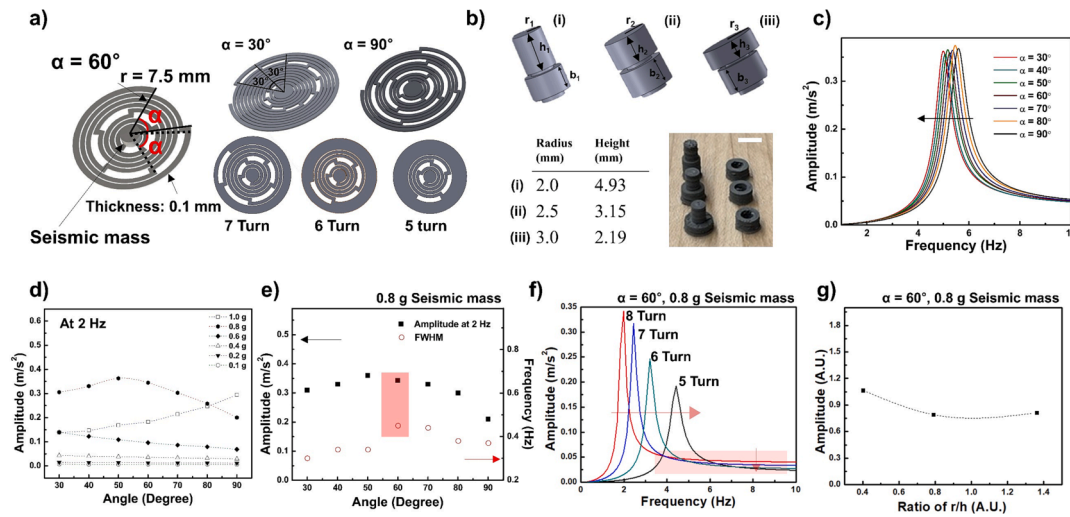


Fig. 3. Shape Programmable Properties of the Selected VK Structure. a) Schematic drawings of shape-programmable properties of the VK structure. The parameters for its shape programming are the α angle to set the repeated cutting slit and the spiral turning number. b) The design of seismic mass has been considered. Its radius and height are varied with fixed volume to be the invariant mass. The length of the b is identical. c) Harmonic responses depending on α angles of the VK structure. d) The responded amplitude at different seismic masses and α angles of VK structure at the vibration frequency of 2 Hz. e) The amplitude, left-black arrow, and FWHM, right-red arrow, depending on α angle at the 2 Hz vibration with 0.8 g seismic mass. f) Response changes of the VK structure with $\alpha = 60^\circ$ and 0.8 g seismic mass at different turning numbers and g) ratio of r/h of the seismic mass. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

different vibration frequencies are investigated. The frequency peak shifting from 6 Hz to 4.5 Hz is observed in the VK structure without the mass from $\alpha = 90^\circ$ to 30° , respectively, in Fig. 3c. The harmonic response peak is closely linked to the structure's natural frequency, which is calculated as $\omega_n = \sqrt{k/m}$. A larger α angle of the VK structure corresponds to a higher frequency at its response peak, signifying an increased ω_n value. Consequently, it can be inferred that a VK structure with a greater α angle leads to a higher spring constant. This suggests that at higher α angles, the connections between each turn of the VK structure exert a more pronounced influence on its spring-like characteristics. This effect occurs even as a higher α angle design results in a reduced overall area and stiffness. This is due to the joint buckling of the kirigami structure in the out-of-plane deformation [21]. After applying seismic mass to the structures and observing their responded amplitude at a fixed 2 Hz vibration, its effectiveness is remarked as shown in Fig. 3d. Overall, the VK structure with $\alpha = 50^\circ$ and 0.8 g mass shows the highest amplitude. As the next step, the FWHM of the responded peak is considered in Fig. 3e. Thus, the VK structure with $\alpha = 60^\circ$ and 0.8 g mass is selected as the reliable structural design for the target application. The graphs related to the harmonic response behavior of the VK structure at different angular designs and amounts of seismic mass are presented in Supplementary Fig. 4. The effect of the turning numbers on the VK structure in Fig. 3f is also studied. A decrease in the number of turns results in the peak shifting towards higher frequency ranges and leads to a decrease in amplitude at 2 Hz. Consequently, we have chosen to utilize 8 turns for the VK structure. However, the rising trends of FWHM at lower turning numbers of the VK structures would be meaningful if the selected criteria outweighed the broader range of vibration frequencies. From the mass design, the effective amplitude changes of the peak are observed as shown in Fig. 3g. A higher ratio between the radius and height of mass design weakens the amplitude of the harmonic response for the VK structure. Thus, mass design (i) is selected.

3.4. The VK vibration sensors

The VK vibration sensors with different seismic masses have been fabricated. A 3D-printed vibration shaker test machine was prepared in-house to evaluate the sensor performance. For the specification of the vibration shaker, details are shown in Supplementary Fig. 5 and Video 2.

The VK sensor is placed on the top bed of the shaker, as demonstrated in Fig. 4a. The sensing performance has been monitored at periodic impact with 0.5 Hz to 2 Hz as shown in Fig. 4b and Supplementary Video 3. The VK vibration sensors with 0.1 ~ 0.2 g seismic masses do not respond to the applied periodic impacts. The signals are observed from the sensors with over 0.4 g mass. The sensors with 0.4 g and 0.6 g seismic masses show identical peak-to-peak values at all applied periodic impacts. However, the sensors with 0.8 g and 1.0 g seismic masses show slightly increasing trends of peak-to-peak values at faster impact periods. 0.8 g mass shows a 27 % increase in the values at 2 Hz periodic impact compared to 0.5 Hz. In contrast, this increase for the VK vibration sensor with 1.0 g mass is 14 %. This sensor shows the highest value at 1.75 Hz, about 30 % increased value, presented in Fig. 4c. The test for harmonic response at imposed vibration with sine waveform has been applied as shown in Supplementary Fig. 6. The consistently responsive trends of the sensor with 0.8 g seismic mass are observed. Thus, the VK vibration sensor with 0.8 g has a reliable performance in detecting the vibration frequency with 2 Hz. Also, the analog signal processes, including three steps low-pass filter, have been used to remove the noise signals from unexpected electromagnetic interferences [43]. The cut-off frequencies are set to be 20 Hz, 10 Hz, and 5 Hz for 1st to 3rd low-pass filters, respectively. The low-pass filtering processes are elaborated on in Supplementary Fig. 6. Although filtered signals show more explicit curves than raw signals, the observed capacitive changes are reduced. Thus, 2nd step filtering method is employed. Furthermore, Fig. 6e presents the enduring sensor signal profile, spanning more than 6,000 cycles of the VK vibration sensors subjected to 1 Hz periodic impacts. The signal profile demonstrates a sustained consistency over time. In Fig. 6f, a marginal discrepancy of 0.017 pF is observed in the maximum capacitance value of the VK vibration sensor after the completion of 6,000 cycles, as compared to its initial state. The sensitivity of the VK vibration sensors with 0.8 g and 1.0 g seismic masses has been calculated as shown in Fig. 4d. Values of 1.95 pF/ms⁻² and 0.80 pF/ms⁻² as the sensitivity are obtained from the VK vibration sensor with 0.8 g and 1.0 g seismic masses, respectively. Although the VK vibration sensor with 1.0 g mass shows higher sensitivity than the sensor with 0.8 g, the standard deviation of ± 0.3 pF of its capacitance changes at different accelerations indicates less reliability than the sensor with 0.8 g mass. The sensitivity of the VK vibration sensor is in the ranges of previously reported

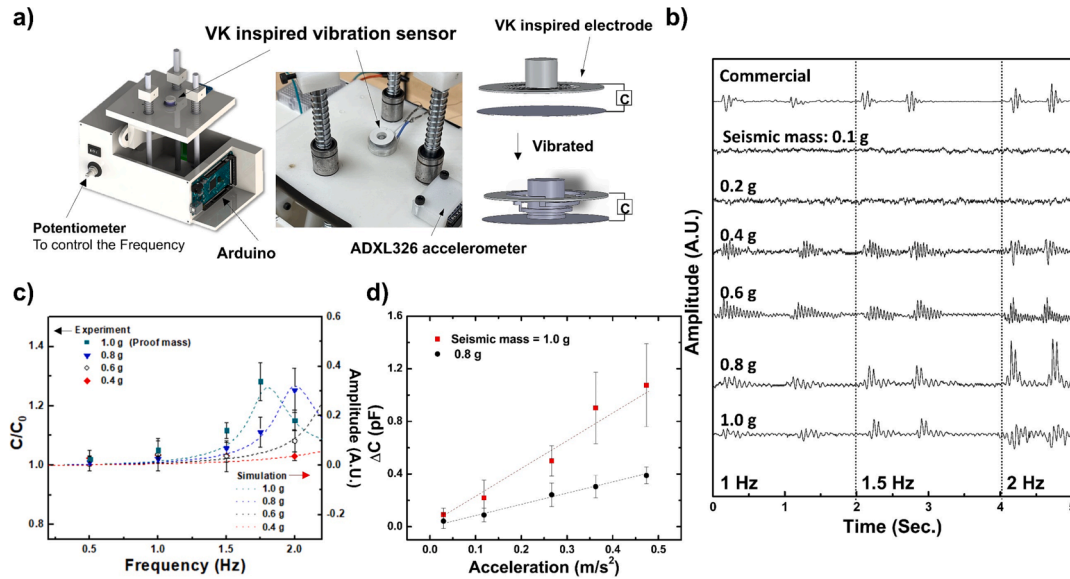


Fig. 4. The VK Vibration Sensors. a) Schematic drawings and actual images of the in-house vibration shaker to evaluate the sensing performance of the VK vibration sensor. The VK electrode is harmonically vibrated when the external vibration is applied to the sensor for the capacitive sensing mechanism. b) Sensor signal profiles of the VK vibration sensors at different seismic masses and imposed periodic impacts. c) A graph for its capacitance changes at different masses and vibration frequencies, symbols with a black arrow, compared to simulation results, dot line with a red arrow. d) Capacitance changes of the VK vibration sensor depending on the applied acceleration. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

performance, $0.124 \sim 1.19 \text{ pF/ms}^{-2}$ as shown in [Supplementary Table 1](#). Most sensors in the table 1 were developed using micro or nano-scale consideration of the materials for the diaphragm. Nonetheless, our case shows highly comparable performance without such consideration, indicating the further potential to improve the sensitivity. Moreover, the sensor with 0.8 g seismic mass performs better in detecting 2 Hz vibration for our target applications. Therefore, the VK vibration sensor

with specific parameters of 0.8 g for seismic mass, 8 for turning number, and 60° for α angle has been chosen for the development of the structural health monitoring system.

3.5. Structural health monitoring of the VK vibration sensing IoT

Structural health monitoring for the internal building site is

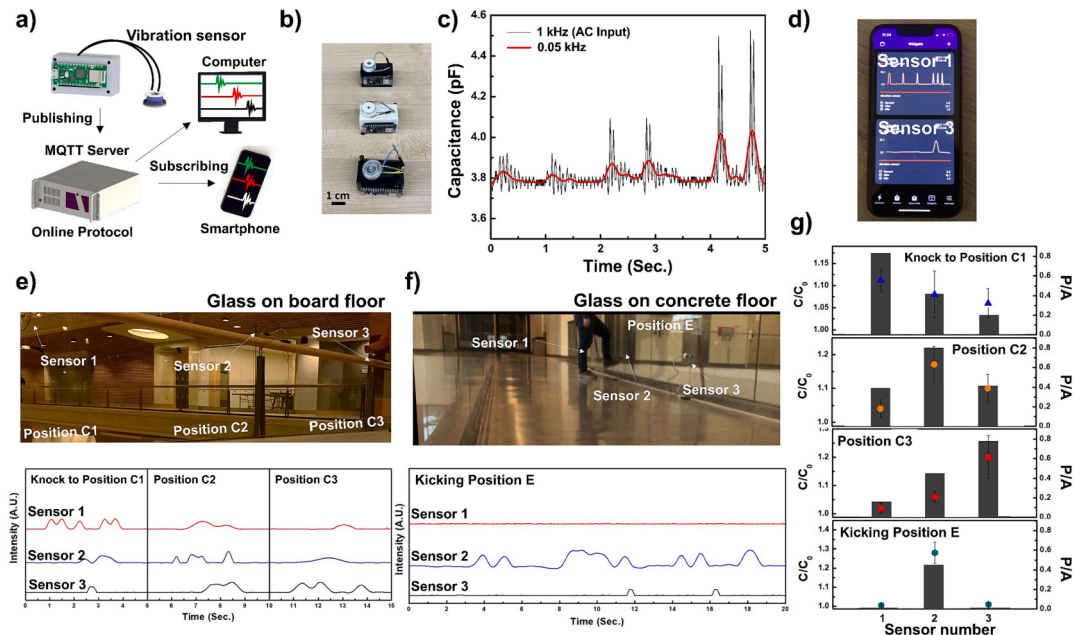


Fig. 5. Structural Health Monitoring of the VK Vibration Sensing IoT. a) Schematic drawings of the portable VK vibration sensing IoT system. When the system is turned on, the obtained vibration data is automatically published to the online MQTT server without any wire connection. Then, the personal computer or smartphone subscribes to the saved data on the server for real-time monitoring. Also, previously recorded data is available to check. b) Actual image of the prepared VK sensing IoT system. The size of the sensing system is $3 \times 2.5 \times 2.5 \text{ cm}^3$. c) The measured vibration signal profiles using the VK sensing IoT system with two different AC inputs. d) Monitoring vibration signal data using smartphone Apps after subscribing to server data. Actual demonstrations of structural health monitoring systems using three VK vibration IoT sensors on the glass wall at the bridge of e) the board floor, f) the concrete floor, and g) their corresponding quantitative values.

important because infrastructure such as bridges or glass walls in the building is directly exposed to people, significantly related to casualties. For example, the glass failure at the bridge of the internal building structure can cause additional accidents for passersby on the lower floor. Such glass failure occurs unintentionally from light impact by passersby which generates multiple or ambient vibrations to affect the infrastructure. Hence, we mainly focused on the vibration caused by tapping, knocking, or walking on the glass wall at the building bridge for the demonstration of our sensor applications. Fig. 5a shows schematics of the structural health monitoring system with the VK vibration sensors through the IoT and MQTT protocols, the VK vibration sensing IoT. It allows us to control and monitor VK vibration sensors anywhere users have WiFi connectable devices such as smartphones or personal laptops [44]. Moreover, monitoring sensing data with multiple devices is available simultaneously because the recorded data are saved in a cloud-based MQTT server as shown in Supplementary Fig. 7a. Thus, combining the portable VK vibration sensors with the IoT system through the MQTT protocol shows the promising feasibility of structural health monitoring. Three portable VK vibration sensing IoT devices with a size of 3 cm × 2.5 cm × 2.5 cm have been prepared as shown in Fig. 5b. Before utilizing them in the structural monitoring system, we check the data generated from the VK vibration sensors. Since the 1 kHz AC input, which generates 1000 data per second, is too fast to be published through the MQTT server at once, post-data processing is required to avoid server errors and missing data. Accordingly, 0.05 kHz AC input for the sensor has been applied to have 50 data per second. However, the issue of potentially missing vibration signals is still there, which can distort the features of vibration profiles. Therefore, as shown in Supplementary Fig. 6, we applied multi-step low-pass filtering. The unwanted noise signal is removed after filtering, and then less data number generated after reducing AC frequency input becomes small enough to show the trends of vibration profiles as shown in Fig. 5c. After completing the VK vibration sensing IoT system, the measured vibration data is transmitted to the MQTT server and is accessible using smartphones, as demonstrated in Fig. 5d. The VK vibration sensing IoTs with external power supplies are also presented in Supplementary Fig. 7b. Their sensing performance at the same position is also demonstrated in Supplementary Video 4. As the next step, three IoT sensing systems are attached to three different glass slides on the bridge in Supplementary Fig. 7c. Fig. 5e shows the IoTs attached to the glasses on the bridge, noted by position C1 ~ C3. A slight knocking shock was exerted on each position of C1 to C3. When knocking was applied to C1, multiple peaks (high-level peak) were detected by sensor 1, while a few minimal peaks were detected by sensor 2 (medium-level peak), placing the position C2 beside the C1. For sensor 3, only one low-level peak was detected. When position C2 between C1 and C3 is knocked, sensors 1 and 3 show medium-level peaks, and sensor 2 shows high-level peaks. These results show that the applied vibration is transferred through the glass wall on the board floor to all IoT sensors. As a comparison, we set three sensors on the glass wall on a vibration-isolated concrete floor in another bridge like Fig. 5f. When the impact is applied to position E, only sensor 2 detects the vibration, and sensors 1 and 3 show minimal vibration peaks. This result indicates that the vibration deliveries through the glass wall are prevented on the concrete floor. The corresponding structural health monitoring with the VK vibration sensing IoTs is demonstrated in Supplementary Videos 5 and 6. Also, their relevant quantitative data are presented in Fig. 5g. Moreover, the number of generated sensor peaks (P) versus the number of actual impacts (A) is calculated as P/A, presented in bar charts. When six times of slight knocking is applied on position C1, sensor 1 generates about five peaks. Thus, the P/A value is 0.8. Simultaneously, about 0.38 and 0.21 as the P/A for sensors 2 and 3, respectively, are obtained. In contrast, in the case of position E on the concrete floor, the P/A values between sensors at the impacted and its adjacent areas show huge differences. These results demonstrate the enhanced properties of the concrete wall when augmented with damping materials [45] for effective vibration isolation [46]. The position E was designed initially with

vibration-proof floors against the movement of heavy objects or crowded people [47]. In contrast, the vibration-proof design of positions C1 to C3 is not considered as much as position E. Such problems may cause the unexpected breakage of the glass wall on the bridge because the occurring vibration and its deliveries [48] through the wall at multiple positions can cause constructive interference for glass failures [49,50]. However, more statistical and time-dependent studies would be supported for a detailed demonstration, which is outside the scope of this study.

4. Conclusion

We developed a 3D-printed vibration sensor inspired by the principles of Vorticella and Kirigami for structural health monitoring. The dimensional morphing of the VK structure enables reliable and reversible deformation in response to targeted vibrations. Additionally, its vibration-responsive behavior is tunable for specific applications and can be produced using multi-material 3D printing in a one-step process without the need for micro- or nano-manufacturing methods or materials. In this study, we investigated the vibration responses from the dimensional morphing of the VK structure and compared them to those of a conventional spring structure. Our results showed that the morphing behavior of the VK structure from 2D to 3D resulted in approximately twice the structural deformation compared to that of the typical spring structure. We also programmed the VK structure to be sensitive at around 2 Hz using its shape-programmable properties. Toward a practical monitoring solution, an Internet of Things (IoT) system has been implemented for the VK sensors to access online sensory data. The VK sensing IoT system enables a smartphone or personal computer to access the measured data without any distance restrictions. In addition, we tested the detection of vibrations delivered through a glass wall at two different bridges. We believe that the demonstrated VK vibration sensing IoT system would be suitable for customized structural health monitoring and highly feasible for any structural sites that require continuous real-time monitoring. Our structural health monitoring system primarily focused on assessing the internal infrastructure of the building, which is typically less influenced by various weather conditions. Consequently, we did not delve extensively into material considerations for fabricating the VK structure in this study. However, for applications involving the building's external structure, a more meticulous material selection process for the VK structure should be conducted as part of our optimization efforts. While we employed a one-step process for creating the VK structure, a more comprehensive perspective reveals our overall approach as a two-step process. This approach to printing in our studies opens up the possibility of using a wider variety of flexible materials to produce intricate structures and associated electronic devices. Nonetheless, this method does come with certain challenges. These challenges include the necessity for recalibration before each subsequent printing step and the requirement for skilled handling to execute the processes effectively. In light of these considerations, we are also exploring the adoption of a one-step process system capable of accommodating multiple materials [39]. However, it's important to note that this particular direction falls outside the scope of our current work. However, the VK design, as it stands, holds the potential to offer a broader array of choices, making it conducive for increased adoption in safety-critical domains.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Woo Soo Kim reports financial support was provided by Natural Sciences and Engineering Research Council of Canada.

Data availability

Data will be made available on request.

Acknowledgments

The authors acknowledge the financial support from the Natural Sciences and Engineering Research Council of Canada (NSERC) with the Discovery and Discovery Accelerator Supplement Grant 2016–04334 and Research Tools and Instruments Grant RTI-2020-00363.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.matdes.2023.112332>.

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